



Acetylene Decomposition



Scenarios Identified

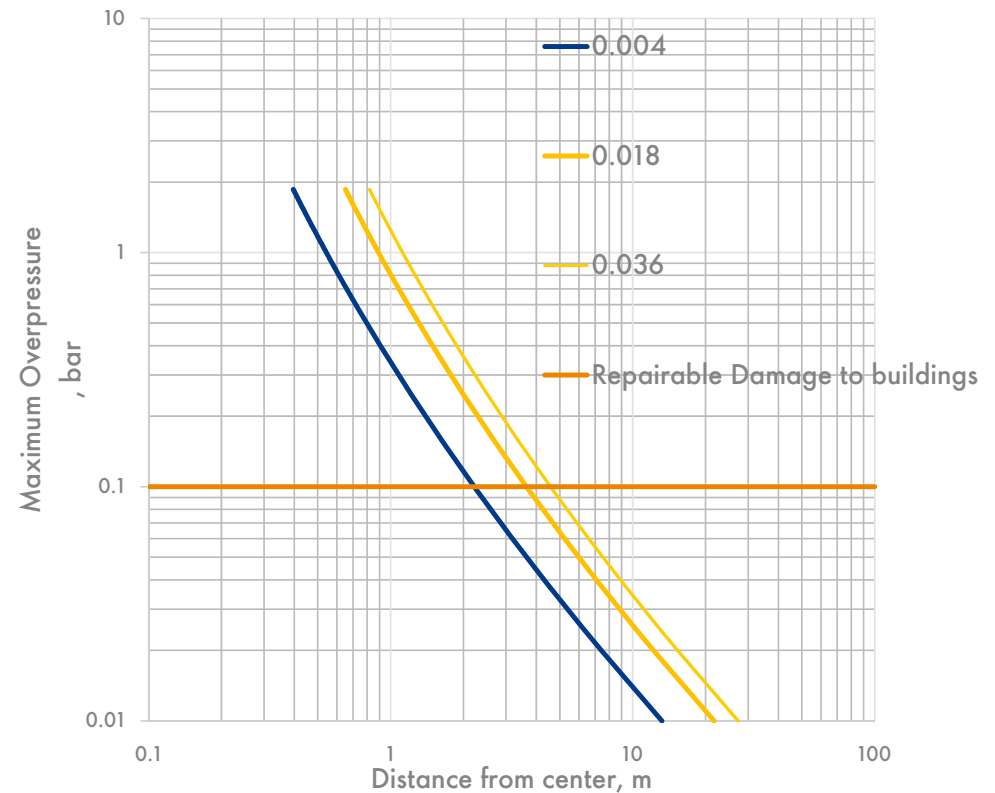
- Thermal decomposition of acetylene
- Internal explosion of acetylene-air mixture

Thermal decomposition of Acetylene

- Heat of reaction 228.3 kJ/mole. The energy content is 8767.3 kJ/kg.
- This decomposition can happen in compressor (500 cm³) or in the discharge KO pot (500 cm³) of the compressor.
- The decomposition is rapid. It is assumed that there is no heat loss to the walls during rapid decomposition. In this case pressure rise will be highest.
- Rupture of the vessel depends in general on rate of pressure rise (dynamic) but also on maximum pressure generated.
- Two approaches are used for the consequence modelling assuming a) that equivalent mass of TNT is available for explosion, b) vessel ruptures maximum pressure.

TNT Explosion

- From the calculations it is estimated that 1 gram of Acetylene is equivalent to 2 grams of equivalent TNT.
- Pressure generation and subsequent decay due to acetylene decomposition.
- The calculation were performed for three different cases a) 2 gram, b) 9 grams and c) 18 grams of acetylene. This corresponds to 4, 18 and 36 grams of TNT.
- Typically we use 70 mbar (0.07 bar) for no fatality due to vapour cloud explosion in case of direct impact of overpressure on people.
- In case of indirect of indirect effects on fatality of people based on building damage, the overpressure is 34 mbar.



Adiabatic Pressure Rise calculation

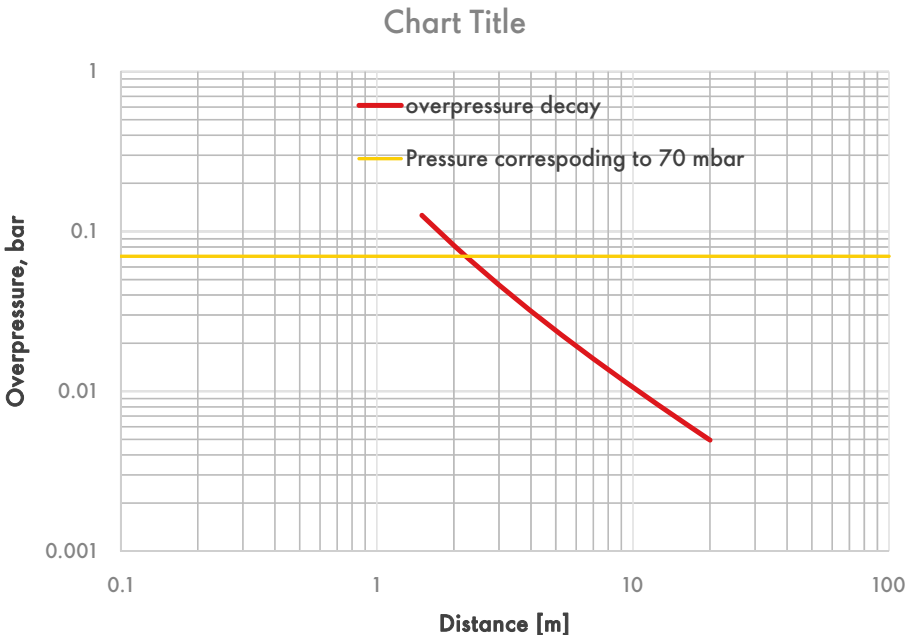
- Based on conservation of energy, final pressure can be calculated.

$$E = \frac{PV}{\gamma - 1} - h_c m_b$$

$$\Delta P = P - P_0 = (\gamma - 1) \frac{h_c m_b}{V}$$

- Here, h_c and m_b is the heat of reaction per kg of acetylene and burned gas available. V is the volume of the enclosure.
- Based on the calculation for 2 grams of acetylene the maximum pressure is estimated to be ~ 80 bar.

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Distance, m	Acetylene Mass in grams			Vessel rupture
	2	9	18	2
Direct effects of explosion (0.07 bar)	2.2	4.5	6	2.5
B3/B4 (0.034 bar)	4.2	7	8.8	4

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Flammability calculation

- The flammability limit of acetylene-air mixture is 2.5 % to 82.0 % volume.
- In case of loss of containment of acetylene (2 grams), it can come in contact with air within the convection oven/skid forming a flammable acetylene-air mixture.
- Volume of air within the convection oven and skid is 1 m³ and 6 m³.
- Based on the flammability calculation it is estimated that the **acetylene-air mixture is not flammable**.
- For the mixture to be flammable acetylene mass should be atleast 30 grams.

Conclusion

- From the flammability calculation it is established that in case of release of acetylene in a skid or in oven, a flammable mixture is not formed leading to no internal explosion.
- For the thermal decomposition, two different calculations have been performed a) assuming acetylene decomposition gives equivalent amount of TNT and b) Rupture of the vessel at maximum pressure generated in an enclosure volume due to thermal decomposition. In the latter it is assumed that there is no heat loss to the vessel walls.
- Note that no credit is taken strength of the vessel. It is assumed that vessel rupture at the corresponding maximum pressure. The rupture of the vessel depends on rate of pressure rise and maximum pressure rise.
- Note that for worst case credible consequence calculation, effects of barriers (e.g. containment of the vessel and design) are not considered.
- Based on the calculation it is estimated that acetylene lab (existing and neighbouring labs) will damage due to the overpressure.
- More accurate calculation of effects of blast pressure on acetylene lab can be performed by using Civil's team.
- Rupture of vessel could lead to generation metal structures or projectiles. FRED has limitation in calculating this accurately.
- It is suggested to do testing of acetylene decomposition and measurement of overpressure generation in Labs in Europe (For example in Poland or in Germany).





Extra slides